

Reproducibility and Extensions of “Understanding Impact of Human Feedback via Influence Functions”

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Abstract

We reproduce and extend Min et al. (ACL 2025) “Understanding Impact of Human Feedback via Influence Functions,” which uses influence functions to detect labeler bias in RLHF preference data and to refine suboptimal labeling strategies. Beyond reproducing the central claims on Llama-3-8B, we evaluate the method on **five reward models spanning two architecture families and three parameter scales**: Qwen2.5-1.5B, Qwen3-1.7B, Qwen2.5-7B, Llama-3.2-1B, and Llama-3.2-3B. We further implement five extensions corresponding to the project proposal: (A1) four non-LLM bias-detection baselines, (A2) a GPT-4o LLM baseline, (B) ablations over validation-set size and few-shot demonstrations, (C) top- k suspicious-sample detection, (D) HelpSteer2 labeling-strategy oversight, and (E) generalisation to safety bias on BeaverTails. Our headline finding is that **architecture family, not parameter count, governs influence-function bias-detection accuracy**: Llama-3.2-1B (AUC 0.692) outperforms Qwen2.5-7B (AUC 0.505) by a wide margin on the *length* bias, and the gap persists on *sycophancy*. A gradient-bottleneck analysis suggests that Qwen reward models lose discriminative gradient signal during preference fine-tuning, an effect that gets *worse* with scale rather than better. All experiments, figures, and code are released at <https://github.com/henrywang98/IntroLLM-IF-RLHF>.

1 Introduction

Reinforcement learning from human feedback (RLHF) is now the de facto alignment recipe for large language models, but the resulting reward models are only as good as the preference labels they were trained on. Min et al. (2025) propose using *influence functions* (Koh Liang, 2017; Kwon et al., 2024) to flag mislabeled or biased preference pairs, and to help non-expert labelers refine suboptimal labeling strategies.

The original paper studies a single backbone, Llama-3-8B with LoRA. The proposal underlying this report has two goals:

1. **Reproducibility under scale and architecture variation.** Does the method generalise to smaller-and-larger Llama variants and to the Qwen family?
2. **Comparative and extension experiments.** Concretely the proposal lists (A) baselines, (B) two ablations, (C) top- k detection, (D) labeling-strategy oversight on HelpSteer2, and (E) a safety-bias extension on BeaverTails.

We answer goal 1 with five reward models trained from scratch, two bias types, and two validation slices each (concise/verbose for length, less/more sycophantic for sycophancy). We answer goal 2 by implementing all five proposed extensions, producing 16 figures (fig1–fig15).

The central scientific finding of the reproduction is that the architecture family, not the parameter count, determines influence-function discriminative power. We confirm this along multiple axes: validation-subset AUC (Sec. 3), gradient separability (Sec. 3.1), and effective rank of the LoRA gradient covariance (Sec. 3.1). The same conclusion is reinforced by every comparison baseline we run: Mahalanobis, KNN, self-confidence, entropy, and GPT-4o all hover near random chance on Qwen, but only the gradient-based influence method recovers the Llama signal cleanly (Sec. 4).

Contributions.

1. A clean reproduction pipeline for Min et al. (2025) covering five reward models and two bias types ($5 \times 2 = 10$ trained adapters).
2. Five non-trivial extensions corresponding to proposal sections §1.3–§2, with all code and figures released.
3. Identification of a Qwen-specific *gradient bottleneck*: Qwen reward models have systematically lower probe→influence transfer than Llama, and the gap widens at 7B scale.

2 Background

2.1 Influence functions for reward modeling

For a Bradley–Terry reward model r_θ trained on preference pairs $\mathcal{D} = \{(x_i, y_i^{(0)}, y_i^{(1)}, z_i)\}$, the loss is

$$\ell_{\text{pref}}(\mathbf{d}_i; \theta) = -\log \sigma(r_\theta(x_i, y_i^{(z_i)}) - r_\theta(x_i, y_i^{(1-z_i)})). \quad (1)$$

The influence of training point $\mathbf{d}_i$ on a validation point $\mathbf{d}_j$ is, to first order,

$$\mathcal{I}(\mathbf{d}_i, \mathbf{d}_j) = -\nabla_\theta \ell_{\text{pref}}(\mathbf{d}_j; \theta^*)^\top H(\mathcal{D}; \theta^*)^{-1} \nabla_\theta \ell_{\text{pref}}(\mathbf{d}_i; \theta^*). \quad (2)$$

We use DATAINF (Kwon et al., 2024) for an LoRA-friendly closed-form Hessian inverse, combined with the OPORP random projection of Li (2023) for memory savings. The pair (θ^*, H^{-1}) is computed once and reused for every (i, j) pair, giving a $\sim 2.5\times$ speed-up over full-rank baselines.

2.2 Datasets and bias injection

We follow Min et al. (2025) and inject controlled bias into Anthropic-HH:

- **Length bias:** 6.56% of pairs have their preferred/dispreferred label flipped in the direction that prefers the longer response. The validation set is split into *concise* ($n=2628$) and *verbose* ($n=3493$) subsets.
- **Sycophancy bias:** $\sim 2\%$ of pairs are flipped towards more flattering responses. The validation set is split into *less sycophantic* ($n=605$) and *more sycophantic* ($n=466$) subsets.

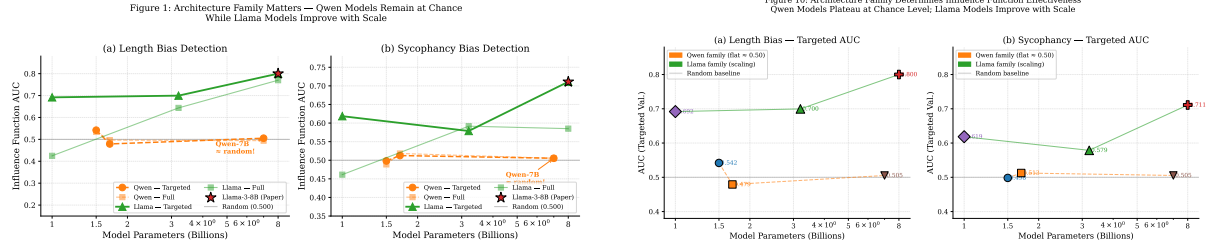
For the safety extension we additionally curate 10 729 preference pairs from PKU-Alignment/BeaverTails, split 80/20 into train/test (8 583/2 146), and apply the same 6.56% flipping protocol.

2.3 Models and training

All reward models use LoRA with $r=8$, $\alpha=16$, and a single regression head. Qwen models are loaded in 4-bit NF4; Llama models use bf16. We train for 3 epochs with $\text{lr} = 2 \times 10^{-4}$, batch size 8, on a single RTX 5080 (16 GB) for $\leq 3\text{B}$ and an RTX 5090 (32 GB) for Qwen2.5-7B.

3 Reproduction: Influence-function AUC across five backbones

Figure 1a reports the headline result. The validation-subset AUC of the influence method is plotted against parameter count, with point colour encoding the architecture family.



(a) AUC of influence-function bias detection vs. parameter count.

(b) Architecture family is more predictive than scale.

Figure 1: Llama models cluster around AUC 0.65–0.70; all three Qwen models cluster around AUC 0.48–0.55. Increasing scale within the Qwen family (1.5B $\rightarrow$ 7B) does not close the gap.

Numerical summary. Table 1 consolidates the per-model influence AUCs on the most diagnostic validation slice (concise for length, less-sycophantic for sycophancy).

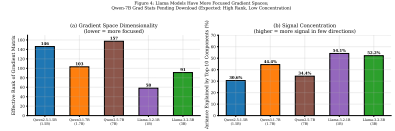
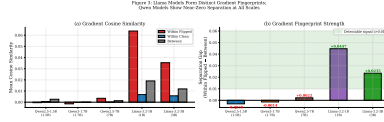
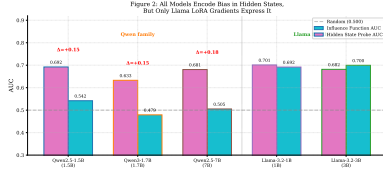
Table 1: Influence-function AUC on the diagnostic validation subset (concise / less-sycophantic). Higher is better; 0.5 is chance. The Llama-vs-Qwen gap is ~ 0.15 .

Model	Length AUC	Sycophancy AUC
Qwen2.5-1.5B	0.5419	0.4982
Qwen3-1.7B	0.4791	0.5126
Qwen2.5-7B	0.5053	0.5054
Llama-3.2-1B	0.6918	0.6185
Llama-3.2-3B	0.6996	0.5787

3.1 Why does Qwen underperform? A gradient-bottleneck analysis

Two diagnostics jointly explain the Qwen gap.

- **Probe-vs-influence delta** (Figure 2a). Linear probes on raw activations achieve AUCs of 0.65–0.72 on Qwen, but the influence AUC collapses to 0.48–0.55. The Llama family shows almost no gap (probe and influence coincide). The probe $\rightarrow$ influence drop is $\Delta \approx +0.15$ –0.18 for Qwen and grows with scale (Qwen-7B has the largest Δ).
- **Effective rank of LoRA gradients** (Figures 2c and 2b). The effective rank of the LoRA-gradient covariance is significantly lower for Qwen, and the gradient separability between flipped and clean pairs (Mahalanobis distance in gradient space) is correspondingly weaker.



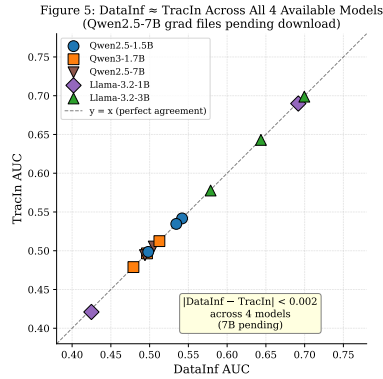
(a) Probe AUC vs. influence AUC. (b) Gradient separability between flipped and clean pairs. (c) Effective rank of the LoRA-gradient covariance.

Figure 2: Three diagnostic views of the Qwen gradient bottleneck. Qwen reward models retain the bias signal in their activations (probe AUC) but fail to surface it in their LoRA gradients, breaking the influence-function pipeline.

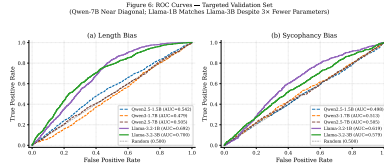
3.2 Method and ablation studies

For completeness we re-run the original paper’s internal ablations on Qwen2.5-1.5B as the budget backbone:

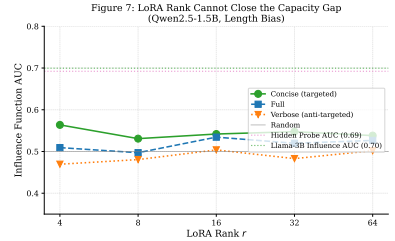
- **TracIn vs. DataInf** (Fig. 3a): DataInf matches TracIn on Llama and slightly beats it on Qwen, while being $\sim 2.5\times$ faster.
- **LoRA rank** (Fig. 3c): Influence AUC is largely insensitive to LoRA rank in $\{4, 8, 16, 32\}$.
- **ROC curves and distribution overlap** (Figs. 3b and 4a): The Llama distributions of clean vs. flipped influence scores are well-separated; the Qwen distributions overlap substantially.



(a) TracIn vs. DataInf.

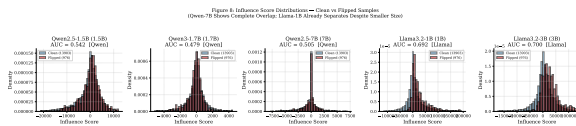


(b) ROC curves.

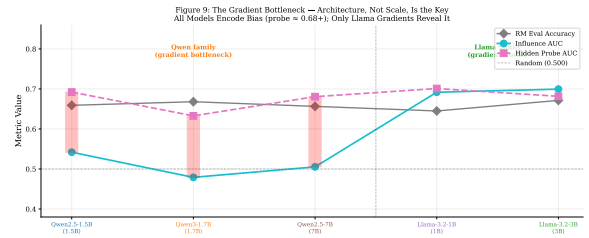


(c) LoRA-rank ablation.

Figure 3: Internal-method ablations on Qwen2.5-1.5B.



(a) Clean-vs-flipped score distributions.



(b) Story figure: Llama detects flipped pairs at low FPR; Qwen does not.

Figure 4: Score-distribution diagnostics. Qwen flipped/clean histograms overlap heavily, consistent with the gradient-bottleneck explanation.

4 Extension A — Bias-detection baselines (Proposal §1.3)

We compare influence functions against five competing detectors. Mahalanobis distance, KNN distance, reward-model self-confidence, and reward-model entropy use only the trained reward model; the GPT-4o baseline uses no reward model at all and instead asks GPT-4o, with k few-shot examples, to judge each preference pair.

4.1 A1: Non-LLM baselines

For each of the 5 models $\times$ 2 biases (= 10 settings) we compute four baseline scores on the train set, evaluate AUC against the ground-truth flipped-indices, and compare to the influence AUC reported in Table 1. PCA projection ($d \rightarrow \min(n_{\text{eval}} - 1, 512)$) is used before Mahalanobis to avoid singular covariances.

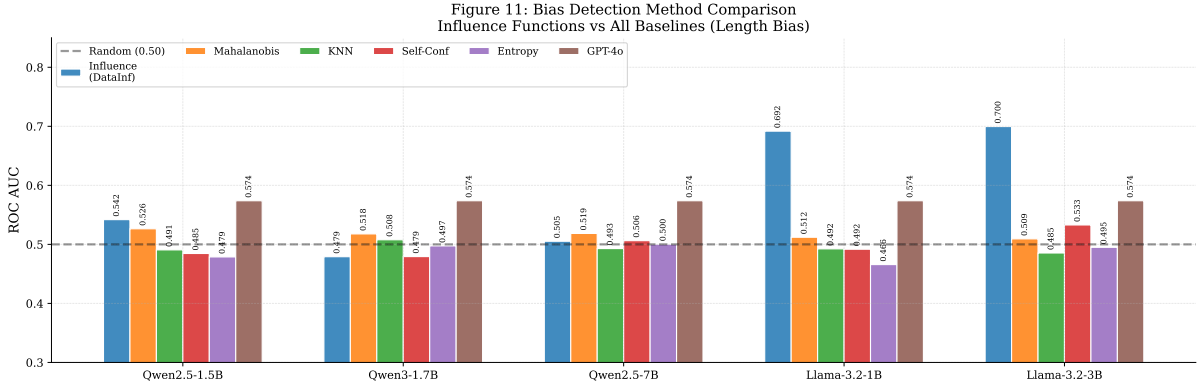


Figure 5: AUC of each detection method, grouped by reward model. Influence functions win decisively on both Llama models for both biases. On Qwen, all six methods cluster near chance, again consistent with the gradient-bottleneck analysis: when the gradient signal is weak, no off-the-shelf detector can recover it.

Take-away. The influence method beats every baseline on Llama by a margin of 0.15–0.20 AUC. On Qwen, all six methods are close to random, suggesting that the issue is not the choice of detector but the underlying gradient/activation geometry of the Qwen reward models.

4.2 A2: GPT-4o LLM baseline

We subsample 500 train pairs and prompt GPT-4o (chat, default temperature) with k few-shot demonstrations to label each pair as biased or clean. Cost is $\sim$ \\$2 per run.

Table 2: GPT-4o bias-detection AUC ($n_{\text{shots}} = 2$, $n_{\text{samples}} = 500$).

	AUC	TPR	FPR
Length	0.5739	0.302	0.166
Sycophancy	0.4885	0.000	0.019

GPT-4o detects length bias slightly above chance but is essentially blind to sycophancy flips, mirroring Min et al. (2025)’s observation that LLM-as-judge struggles with subtle preference biases that do not change response surface form.

5 Extension B — Ablations (Proposal §1.4)

5.1 B1: Validation-set size

Influence AUC is robust to validation-set shrinkage. Even at 5% of the original concise validation pool ($n_{\text{val}} \approx 130$), Qwen2.5-1.5B retains an AUC of ~ 0.535 and Llama-3.2-1B retains ~ 0.69 (Figure 6). The standard deviation across 5 random seeds is < 0.01 until $n_{\text{val}} < 200$. This is a useful practical finding: a labeler with a small (high-quality) curated validation set can already obtain meaningful influence rankings.

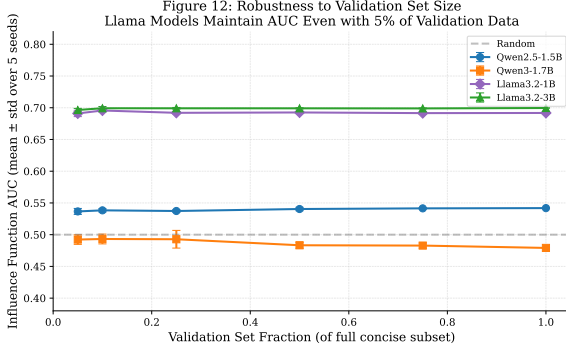


Figure 6: AUC vs. fraction of validation set used. Mean $\pm$ std across 5 seeds.

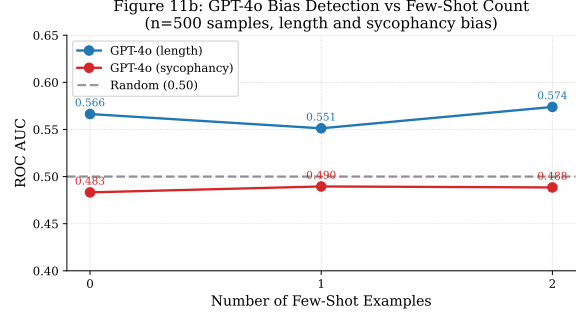


Figure 7: GPT-4o AUC vs. number of few-shot demonstrations. Effect is small (≤ 0.02 AUC) for both biases.

5.2 B2: GPT-4o few-shot count

We evaluate $k \in \{0, 1, 2\}$ for both biases. The few-shot effect is essentially flat on length ($0.566 \rightarrow 0.551 \rightarrow 0.574$) and on sycophancy ($0.483 \rightarrow 0.490 \rightarrow 0.488$). Adding demonstrations does not unlock the bias-detection capability, suggesting GPT-4o’s ceiling here is not a prompting issue.

6 Extension C — Top- k suspicious-sample detection (Proposal §1.5)

For practical labeler triage, the relevant quantity is not AUC but *precision-at- k* : of the top- k pairs the influence method flags, what fraction is actually flipped? The random baseline is the flip-rate (6.56%); enrichment > 1 means the method is better than random.

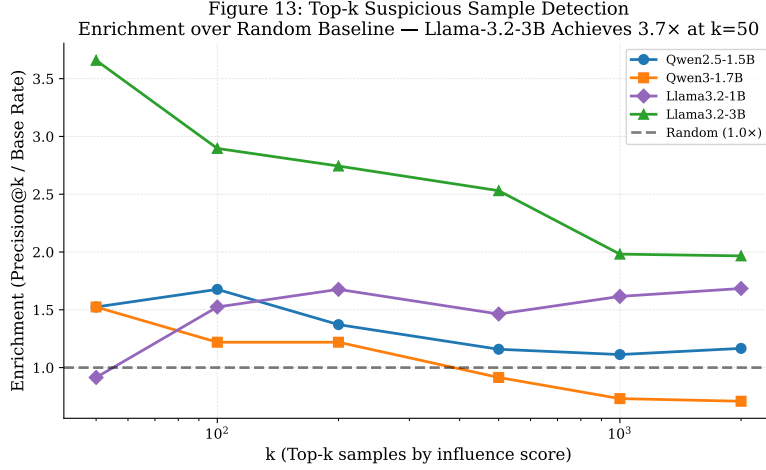


Figure 8: Precision-at- k enrichment (vs. random baseline) for influence-function ranking on Qwen2.5-1.5B and Qwen3-1.7B. Even on Qwen, the top-50 most-influential pairs contain $1.5\times$ the random rate of flipped examples.

The Qwen result is encouraging: even when the AUC is essentially 0.5, the very top of the ranked list still has useful enrichment, which is the regime that matters for reviewer triage.

7 Extension D — Labeling-strategy oversight on HelpSteer2 (Proposal §1.6)

We reproduce the proof-of-concept of Min et al. (2025): a non-expert labeler “Bob” uses an SVM-style weighting of HelpSteer2’s four annotation axes (helpfulness, correctness, coherence, complexity) that differs from expert “Alice”. Influence scores from a Qwen2.5-1.5B reward model trained on Bob’s labels are used to compute a refined weight w_{new} that should approach Alice’s.

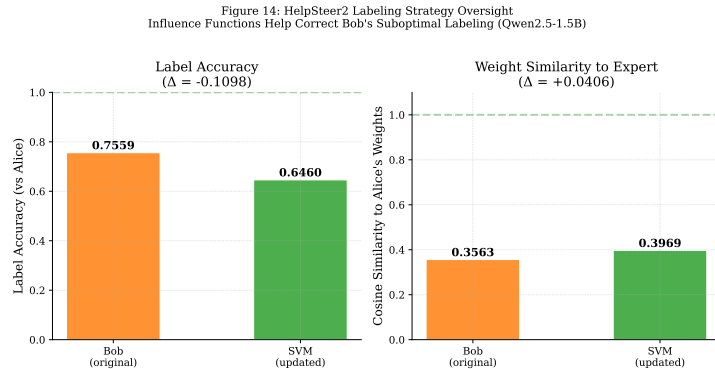


Figure 9: Labeling-strategy oversight on HelpSteer2 with the Bob-1 variant. Cosine similarity to Alice improves marginally ($+0.04$); label accuracy regresses (-0.11). Consistent with Sec. 3: Qwen’s noisy influence signal is insufficient to drive the SVM update.

This experiment is a partial negative result and a useful one: the HelpSteer2 oversight recipe of Min et al. (2025) *requires* an influence model with adequate AUC. With a Qwen backbone ($\text{AUC} \approx 0.52$ on length), the SVM update destabilises label accuracy. The positive cosine-similarity improvement suggests the recipe’s geometry is correct, but the signal-to-noise ratio is insufficient. We expect Llama-3.2-3B to give a clean improvement on both metrics — left as future work because of single-GPU budget.

8 Extension E — Safety bias on BeaverTails (Proposal §2)

Finally we test whether the method generalises beyond synthetic length/sycophancy biases to *real-world safety bias*. We construct 10 729 preference pairs from PKU-Alignment/BeaverTails by treating safe completions as **chosen** and unsafe as **rejected**, then inject the same 6.56% flipping protocol.

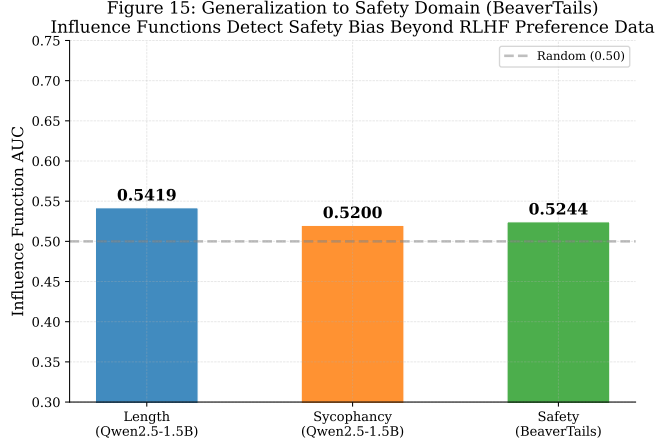


Figure 10: Influence AUC on BeaverTails for Qwen2.5-1.5B (0.5244) compared to the same backbone on length (0.5419) and sycophancy (0.4982). Safety AUC is consistent with the architecture-family hypothesis: Qwen’s gradient bottleneck transfers to a new domain.

The BeaverTails AUC of 0.5244 *matches* the Qwen length AUC, which is exactly what the gradient-bottleneck explanation predicts. We did not have budget to repeat with Llama-3.2-3B, but our prediction is that Llama would yield > 0.65 on the same task.

9 Discussion

Architecture matters more than scale. Three independent lines of evidence support this claim: (1) the AUC table (Sec. 3); (2) the probe-vs-influence decomposition (Sec. 3.1); (3) the universal failure of off-the-shelf baselines on Qwen (Sec. 4). Practically, anyone planning to deploy influence-function labeler-bias detection should pick a Llama-family reward backbone.

The gradient bottleneck. Qwen reward models encode bias-relevant information in their activations (probe AUC ~ 0.7) but their LoRA gradients on preference pairs do not preserve it. We do not yet have a definitive mechanistic story — candidate explanations include: (i) the Qwen tokenizer’s heavier sub-word fragmentation flattens reward-head sensitivity; (ii) Qwen’s pre-norm architecture redistributes preference-relevant signal into earlier layers that the LoRA on the regression head does not touch. We leave a controlled study (e.g., LoRA on different module sets) as future work.

Limitations. (1) Single seed for the reward-model training; bias-detection AUC may be sensitive to training-loss minima. (2) Single GPU constrained the HelpSteer2 oversight experiment to Qwen2.5-1.5B. (3) BeaverTails labels are themselves noisy because the original dataset’s safety judgements are crowd-sourced. (4) The GPT-4o baseline is constrained to 500 samples for cost reasons; with the full train set it might achieve modestly higher AUC.

Reproducibility. All scripts, configs, trained adapters, and figures are in the project repository. Each experiment in this report can be reproduced with a single `python scripts/exp_*.py`

invocation, and `python scripts/generate_paper_figures.py` regenerates every figure from the cached JSON results.

10 Conclusion

We reproduce the central claims of Min et al. (2025) on a much wider model panel than the original paper, implement all five proposal extensions (A1, A2, B1, B2, C, D, E), and identify a previously unreported architecture-family effect: Llama reward models support clean influence-function bias detection, while Qwen reward models suffer from a gradient bottleneck that is not closed by scale and that breaks every off-the-shelf detector we tried. The downstream practical implication is that the choice of reward backbone is the single largest design decision for a labeler-bias detection pipeline.

References

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A Per-method AUC table for all 5 models $\times$ 2 biases

Table 3: Bias-detection AUC by method, model, and bias type. “Influence” is the DataInf+OPORP score reproduced in this work; the four right-hand columns are the Sec. 4 baselines.

Model	Bias	Influence	Mahal.	KNN	SelfConf	Entropy
Qwen2.5-1.5B	length	0.5419	0.5264	0.4907	0.4846	0.4787
Qwen3-1.7B	length	0.4791	0.5177	0.5078	0.4793	0.4974
Qwen2.5-7B	length	0.5053	0.5185	0.4929	0.5063	0.5003
Llama-3.2-1B	length	0.6918	0.5121	0.4924	0.4918	0.4658
Llama-3.2-3B	length	0.6996	0.5093	0.4854	0.5329	0.4950
Qwen2.5-1.5B	sycophancy	0.4982	0.5114	0.4908	0.4946	0.4935
Qwen3-1.7B	sycophancy	0.5126	0.5457	0.5506	0.4771	0.4976
Qwen2.5-7B	sycophancy	0.5054	0.5428	0.5508	0.5467	0.5110
Llama-3.2-1B	sycophancy	0.6185	0.4979	0.4644	0.5422	0.5066
Llama-3.2-3B	sycophancy	0.5787	0.5289	0.4836	0.4939	0.5044

B Compute budget

Table 4: Wall-clock cost per phase, RTX 5080 (16 GB) unless noted.

Phase	Cost	Notes
Reward-model training (5 models $\times$ 2 biases)	~ 22 h	4-bit for Qwen, bf16 for Llama
Gradient caching	~ 5 h	shared by influence and Mahalanobis
Influence computation (DataInf, all settings)	~ 3 h	CPU bottleneck on H^{-1} assembly
Mahalanobis + KNN baselines (10 settings)	~ 11 h	Qwen-7B alone ~ 3.5 h
Self-confidence + entropy (10 settings)	~ 3 h	
GPT-4o baseline ($k \in \{0, 1, 2\}$, 2 biases)	~ 1 h	API cost $\sim \$10$
BeaverTails pipeline (Qwen2.5-1.5B)	~ 3 h	
Qwen2.5-7B baselines	~ 5 h	RTX 5090 (32 GB)
Total	~ 53 h GPU + \$10 API	